

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Caribbean ED Validation of Machine-Learning Triage Support

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Statement of the Problem:

Most AI-based triage tools have been developed and tested outside the Caribbean, making it unclear whether they would perform effectively within local emergency department environments. (1–6) At Mercer General Hospital, differences in workflow, staffing, resources, and patient populations may affect the safety and practicality of AI-supported triage. (7,8) Therefore, the problem is determining whether a machine-learning triage support tool can be validated and integrated into the hospital's emergency department workflow while improving identification of high-risk patients without increasing staff workload or compromising patient safety. (2,8,9)

Previous Work:

Tyler et al. examined how artificial intelligence and machine learning are being used in emergency department triage. The problem addressed was that conventional triage systems may struggle with risk prediction, patient prioritisation, and resource planning in busy emergency departments. Using a scoping review method, the authors found that AI and machine learning models were often used to predict hospital admission, disease risk, resource needs, and patient outcomes. However, many studies relied on retrospective data, single-hospital datasets, and limited long-term outcome evidence, which reduces confidence in applying the findings widely.(1)

Da'Costa et al. reviewed the role of AI-driven triage systems in emergency departments affected by overcrowding, limited resources, and inconsistent patient prioritisation. Their method was a literature review discussing the benefits, challenges, and future direction of AI triage. The authors found that AI may reduce waiting times, support clinical decision-making, improve patient flow, and assist with resource allocation. However, they highlighted challenges such as poor data quality, algorithmic bias, privacy concerns, accountability issues, limited clinician trust, and the need for more diverse datasets.(2)

Araouchi and Adda developed TriageIntelli, an AI-assisted multimodal triage system for health centres using the Korean Triage and Acuity Scale. Their study compared several machine learning models, including Support Vector Machine, Random Forest, Artificial Neural Network, Gradient Boosting Machine, Logistic Regression, XGBoost, and a stacking model. The stacking model performed best, with an accuracy of 80.05%. However, the authors noted that the model relied on a single dataset, which limits how confidently the results can be applied to other hospitals, populations, and healthcare systems.(3)

Kang et al. developed a predictive model to assign emergency department patients to appropriate care zones using data from Sengkang General Hospital in Singapore. The model used routinely collected triage information such as vital signs, demographics, NEWS scores, and nurses' triage notes. The Light Gradient-Boosting Machine model performed best, with strong AUC results during both holdout testing and later clinical validation. However, the study was based on local

hospital data, meaning further testing would be needed before applying the model in other settings such as Caribbean emergency departments.(4)

Alomari et al. examined whether ChatGPT could support emergency department triage using the Canadian Triage and Acuity Scale. The study found high agreement between ChatGPT and emergency physicians, but much lower accuracy when compared with a senior emergency department consultant. This suggests that general AI tools may be useful for triage support, but they are not reliable enough to replace trained clinical judgement. The authors also noted limitations such as small sample size, case complexity, missing information, and the fact that the AI model was not specifically trained for medical triage.(9)

Wang et al. studied whether large language models could support patient triage and guide patients to the correct department. After prompt engineering, GPT-4-Turbo achieved very high MEWS-based triage accuracy, while GPT-4o also performed strongly for triage and outpatient department guidance. The findings suggest that large language models may have potential as decision-support tools in emergency care. However, the study used Chinese patient scenarios, structured descriptions, and Chinese hospital department categories, so the results may not fully transfer to other health systems or Caribbean emergency departments. (5)

De Freitas et al. explored patient flow in a Caribbean emergency department, which is important because ED delays and workflow barriers can affect patient care. The study used qualitative methods, including non-participant observation, process mapping, and informal conversational interviews. The authors found that patient flow was influenced by work processes, ED layout, material resources, nursing staff levels and roles, non-clinical staff, and links with other hospital departments. Although the study was not about AI triage, its limitation and value for this proposal is that it provides local Caribbean workflow context, showing why any AI triage support tool must be validated within the setting where it will be used.(7)

Cho et al. tested a real-time medical record input assistance system using voice artificial intelligence for emergency department triage tasks. The AI system reduced the median time needed to complete triage documentation compared with manual input, showing that AI may improve the speed of triage workflow. However, manual input still had higher completion rates for several variables, and the AI system had lower accuracy with categorical information such as past medical history and allergy history. This paper is useful because it shows that AI can support triage workflow, but also that technical reliability and data quality must be addressed before replacing conventional methods. (8)

Bavali-Gazik et al. developed and validated an AI-based electronic triage model for cardiac-suspected patients in the emergency department. The study compared models using routine triage data, ECG interpretation, and a combined fusion model. The fusion model performed best and outperformed the conventional Emergency Severity Index, suggesting that AI can help identify high-risk patients when routine triage data are combined with other clinical information. However, the study was retrospective, cardiac-specific, and based on one hospital setting, so its findings still need broader validation before being applied in other emergency departments. (6)

Gaps Identified:

Two main gaps were identified from the literature. The first gap is limited validation in Caribbean emergency department settings. Several studies reported strong AI triage performance, but many were tested in one country, one hospital, one patient group, or one triage system. This means Mercer General Hospital cannot assume that an AI model developed elsewhere would perform safely, fairly, or accurately for Caribbean patients, hospital resources, and emergency department workflows. The Caribbean patient-flow study also shows that local ED processes, staffing,

resources, and hospital links can strongly affect how any new support tool would work in practice. (3–7)

The second gap is limited evidence on practical workflow integration. Many AI triage studies focus mainly on model accuracy, but fewer explain how the tool would fit into the real triage process without slowing staff down, increasing documentation burden, or reducing data quality. This is important because an AI tool should support nurses and doctors, not replace clinical judgement or create extra work during an already busy triage process. For Mercer General Hospital, the proposed tool should therefore be tested as a support layer beside existing triage methods, with attention to high-risk patient identification, over-triage, staff usability, and workflow fit. (1,2,8,9)

Proposed Solution:

To address the lack of Caribbean validation and the limited evidence on workflow integration, this project proposes a 12-week pilot study to test a machine-learning triage support tool using de-identified emergency department data from Mercer General Hospital. The tool would not replace nurses, doctors, or existing triage systems. Instead, it would act as a support layer beside current triage practice, helping staff identify patients who may be at higher risk while keeping the process familiar, practical, and clinically supervised. (2,7,8)

The pilot would use routinely collected triage information such as age, sex, vital signs, chief complaint, mode of arrival if available, and existing triage category. A simple baseline model, such as logistic regression or a decision tree, would be developed first because these models are easier to interpret and explain. If time allows, the baseline model could then be compared with stronger machine-learning models such as Random Forest or Gradient Boosting to assess whether more complex methods improve performance enough to justify their use. (3,4,6)

The model would be evaluated using clinically relevant measures rather than accuracy alone. These would include high-acuity recall, under-triage risk, over-triage rate, agreement with existing triage outcomes, and the model's ability to identify patients who may need urgent review, admission, or higher-level care. This is important because an AI triage tool must prioritise patient safety and should reduce the risk of missing high-risk patients while avoiding unnecessary pressure on already limited emergency department resources. (1,2,9)

A simple workflow mock-up would also be created to show how the tool could fit into Mercer General Hospital's existing triage process. The mock-up would show where a risk score or alert could appear, how nurses and doctors might interpret it, and how the tool could support decision-making without creating extra documentation work. This workflow focus is important because Caribbean emergency department performance is affected by local staffing, resources, layout, work processes, and links with other hospital departments. (7,8)

The pilot would remain within a safe 12-week scope. It would use only de-identified data and would not involve live hospital deployment, autonomous decision-making, regulatory approval, or direct changes to patient care. Any future clinical use would require stronger validation, clinical oversight, stakeholder feedback, and hospital or ethics approval.

References (5 to 10 *peer-reviewed journal or conference papers*):

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